

Exploring the Sonic Universe of the Sega Genesis

Data Science, FM Synthesis, and 93,000+ Presets

Abraham Casas Maria Cerro



Ludo2026 — Session 9: *Synthesis, Served Three Ways*

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Who we are



Abraham Casas

Electronics & CTO

Presenting today



DAFM SYNTH ARCADE, GENESIS &
BLASTER



Maria Cerro

Design & CEO

Kasser Synths

We bring three Yamaha FM worlds back into musicians' hands: the **YM2151** (OPM) of the arcades, the **YM2612/YM3438** (OPN) of the Mega Drive / Genesis, and the **YMF262** (OPL3) that coloured 1990s Sound Blaster PCs. **Today we focus on the Genesis.**

Sega's chip bet — YM2612 + SN76489



YM2612 / YM3438 (OPN2)

Genesis audio follows arcade lineage (e.g. System 16):
68000 + Z80 + Yamaha FM — scaled for the living room.

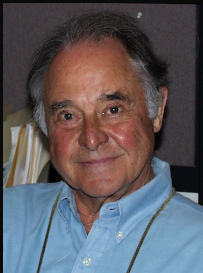
- ▶ **YM2612 / YM3438** — 6-ch. 4-op FM
- ▶ **SN76489** — SG-1000 / Mark III / Master System PSG

"We already used the Yamaha FM sound chip for arcade games, while the TI sound generator was employed to retain SG-1000 and Mark III compatibility."

— Masami Ishikawa (Mega Drive hardware), Polygon interview, 2015

OPN2 is **console-scale** OPN (not YM2151 / YM2608):
onboard DAC, fewer extras. Trade-off: "low-volume"
grit.

What is FM? — Chowning's discovery



John Chowning / Stanford



Yamaha DX7 (1983)

FM in one sentence:

one oscillator changes another's *frequency* so fast that we hear new **timbre**, not vibrato.

1967: ear discovery first; maths later.

Stanford patents it — one of its **most lucrative** licenses (~\$20M+ to Yamaha).

U.S. organ makers pass;
Yamaha builds the DX7.

Sources: Yamaha Hub, "Discovering Digital FM: John Chowning Remembers" (Kovarsky, 2020); Stanford OTL — FM among the university's top royalty generators.

Anatomy of a FM synthesis preset

Use case: YM2612 / YM3438 (OPN2 / OPN2C)

A preset is one **channel recipe**: how four operators are wired, levelled, and enveloped.

Channel: ALG/CON (wiring), **FB**, L/R, AMS/PMS.

Each op: DT/MUL, TL, envelope (KS, AR, DR, SR, SL, RR), AM, SSG-EG.

Hear the idea:

► **Carrier** almost sine

► **FM** op→op

► **AM** LFO tremolo

► **FB** feedback

ALG chooses who modulates whom.

M = modulator, C = carrier (audible).

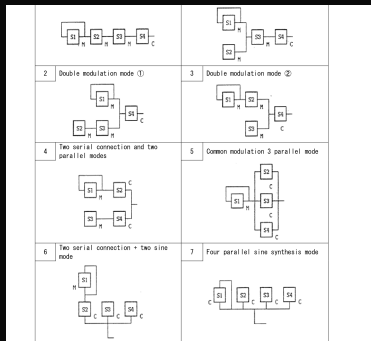


Figure 2-3: Algorithm

M: Modulator
C: Carrier

DT / MUL and TL — pitch class & mix level

DT (Detune) + MUL (Multiple)

- ▶ Each operator is a sine at a harmonic ratio of the note
- ▶ **MUL** = 0 is a special half-ratio; 1 = fundamental; 2, 3, ... = overtones
- ▶ **DT** fine-shifts that ratio (chorus / inharmonic edge)
- ▶ Modulator MUL → which sidebands; carrier MUL → the pitch you hear

TL (Total Level) — 0 loud ... 127 silent

- ▶ **Carrier TL** $\approx$ loudness in the mix
- ▶ **Modulator TL** $\approx$ FM richness (more sidebands)

Rule of thumb

Carrier TL = volume fader.

Modulator TL = FM index (brighter, not just louder).

Later TL-dedup: same wiring + envelopes, different TL $\approx$ same instrument at another mix level.

Envelopes — KS, AR, DR, SR, SL, RR (+ AM)

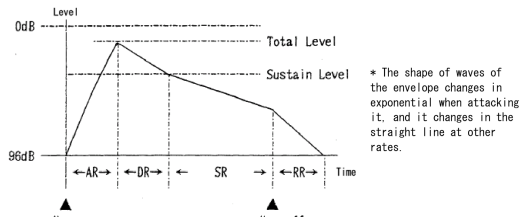
Each operator has its own envelope (not one ADSR for the whole patch).

Param	What it does
KS	Speeds envelope on higher notes
AR	Rise from silence (31 = instant)
DR	Fall from peak toward sustain
SL	Where that first decay lands
SR	Drift while held (0 = freeze)
RR	Fall after key-off
AM	Op level breathes with the LFO

Musical picture

Pluck = fast AR + fast modulator decays; pad = slow AR on carriers. Modulator envelopes → *colour*; carriers → *loudness*.

is shown by each rate and sustain level of decay, sustain, and release. Figure 2-4 shows the envelope wave form and each parameter.



OPN envelope (AR→DR→SR→RR); same family as YM2612.

SSG-EG, FB / ALG, L/R, AMS / PMS

SSG-EG

Legacy envelope shapes (repeating / alternate). On YM2612 game patches it is rarely the star — know it exists.

FB + ALG (FL + CON)

ALG = wiring (carriers vs modulators). **FB** feeds op. 1 into itself — grit and cut.

► **FB** max feedback

L / R

Stereo enable for the channel.

AMS / PMS + LFO

LFO speed; **AMS** = amplitude depth; **PMS** = pitch depth. Per-op **AM** bit opts into AMS — tremolo without new FM sidebands.

► **AM** AMS tremolo

VGM is not a recording — it is the chip's command stream

A VGM file logs, sample-accurately, the register writes the game sent to the sound hardware — not a rendered WAV.

- ▶ Replay = hear the music as the chip was programmed mid-song
- ▶ For Genesis: YM2612 ports (e.g. commands 0x52/0x53)
- ▶ Format born on **SMS Power!** (Maxim); expanded across chips on **VGMrips**
- ▶ **Project2612**: quality-controlled Genesis / Mega Drive archive

The format starts with a 192 byte header:

	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
0x00	["Vgm " ident]	[EoF offset]	[Version]	[SN76489 clock]												
0x10	[YM2413 clock]	[GD3 offset]	[Total # samples]	[Loop offset]												
0x20	[Loop # samples]	[Rate]	[SN FB]	[SNW][SF]	[YM2612 clock]											
0x30	[YM2151 clock]	[VGM data offset]	[Sega PCM clock]	[SPCM Interface]												
0x40	[RF5C68 clock]	[YM2203 clock]	[YM2608 clock]	[YM2610/B clock]												
0x50	[YM3812 clock]	[YM3526 clock]	[Y8950 clock]	[YMF262 clock]												
0x60	[YMF278B clock]	[YMF271 clock]	[YM2280B clock]	[RF5C164 clock]												
0x70	[PWM clock]	[AY8910 clock]	[AYT][AY Flags]	[VM] *** [LB][LM]												
0x80	[GB DMG clock]	[NES APU clock]	[MultiPCM clock]	[uPD7759 clock]												
0x90	[OKIM6258 clock]	[OF][KF][CF] ***	[OKIM6295 clock]	[K051649 clock]												
0xA0	[K054539 clock]	[HuC6280 clock]	[C140 clock]	[K053260 clock]												
0xB0	[Pokey clock]	[QSound clock]	*** **	*** **												

Header clocks many chips (YM2612 among them).

Spec, players, and tools are living community infrastructure — not a one-off dump.

Community pipeline — log, clean, then extract

Three layers of unpaid craft sit between a cartridge and an OPM bank.

Stage	What happens	Heroes / tools
1. Log	Emulator records every sound-chip write while the game runs	BlastEm (Mike Pavone; today's Genesis logger); historically Kega Fusion . Format: Dave + Maxim (SMS Power!, 2001)
2. Clean	Trim loops, find loop points, tag GD3, optimise / compress	Valley Bell's vgmtools: vgm_trim, vgm1pfnd, vgm_tag, optvgm (Blargg→VB), vgm_cmp. Countless rippers + Project2612 (Dark Pulse, DJ Squarewave, DukeNukem)
3. Extract	Lift FM channel configs out of the command stream	Shiru vgm2opm → OPM (what DrWashington batch-ran). Later: NeoLogiX vgm2pre (TFI/VGI/...)

Without these people there is no corpus — only ROMs.

Registers $\Rightarrow$ the composers' FM presets

Because the log contains the same memory the chip used, the voice recipes are inside the stream — ALG, FB, TL, envelopes, MUL...

- 1 Game music runs $\rightarrow$ CPU writes YM2612 registers
- 2 Emulator logs those writes as **VGM** (stage 1)
- 3 Community trims / tags / optimises (stage 2)
- 4 **vgm2opm** (Shiru) dumps channel voices to **OPM** (stage 3)

20bit A*5x-9	CH 1 / CH 4				CH 2 / CH 5				CH 3 / CH 6			
	S1	S3	S2	S4	S1	S3	S2	S4	S1	S3	S2	S4
DT/MULT1	30	34	38	3C	31	35	39	3D	32	36	3A	3E
TL	40	44	48	4C	41	45	49	4D	42	46	4A	4E
KS/AR	50	54	58	5C	51	55	59	5D	52	56	5A	5E
AM/DR	60	64	68	6C	61	65	69	6D	62	66	6A	6E
SR	70	74	78	7C	71	75	79	7D	72	76	7A	7E
SL/RR	80	84	88	8C	81	85	89	8D	82	86	8A	8E
SSG-EG	90	94	98	9C	91	95	99	9D	92	96	9A	9E
F-Num1	A0				A1				A2			
Block/F-Num2	A4				A5				A6			
#F-Num1 * 1									A9	AB	AA	A2
#Block/F-Num2 * 1									AD	AC	AE	A6
FB/Algorithm	B0				B1				B2			
L/R,AMS/PMS	B4				B5				B6			

OPM = portable voice dump (VOPM / our analysis).

The underexplored archive — DrWashington's dump

KVR Audio, Feb 2010 — <https://www.kvraudio.com/forum/viewtopic.php?t=277864>

DrWashington sat on top of that pipeline: he ran **all of Project2612** through Shiru's **vgm2opm** and posted the banks for VOPM (plus SysEx for FM8):

"I just converted pretty much every Sega Genesis game ever into .opm files ... There are literally thousands of useful FM sounds ... from mostly Japanese FM wizards in the early '90s."

- ▶ Seed of our **93,000+** voice corpus — the source dataset for this study



Data extraction — Making the archive speak

Turning raw OPM into an analysis-ready corpus:

- 1 Parse presets (channel + 4 operators $\approx$ 58 parameters)
- 2 Deduplicate by timbre (ignore **TL** volume variants)
- 3 Normalise game names (regions, aliases)
- 4 Enrich: composer, nationality, **GEMS**

OPM file structure

@:0 Preset Name

LF0: 0 0 0 0 0

CH: 0 0 0 0 0 0 0

M1: 31 0 0 0 0 0 0 0 0 0 0

C1: 31 0 0 0 0 0 0 0 0 0 0

M2: 31 0 0 0 0 0 0 0 0 0 0

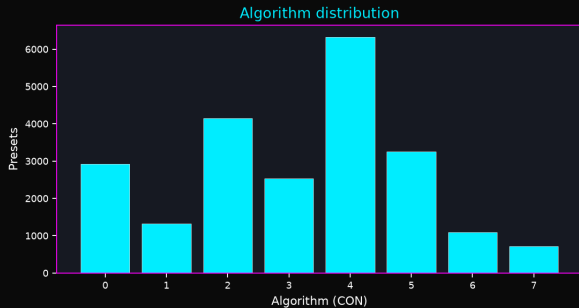
C2: 31 0 0 0 0 0 0 0 0 0 0

One voice = name + LFO + channel (**CH**) + four ops (**M1**, **C1**, **M2**, **C2**).

OPM gives chip parameters. Context (who, which tool, which region) comes from public sources we join in.

[illegible]

Data Analysis: Algorithm 4 as the workhorse



- ▶ **Algorithm 4** is the corpus default: two modulated carriers summed — two FM voices in one channel
- ▶ Composer-friendly: wide palette, easy to balance in a busy mix
- ▶ Other strong CONs also make musical sense (next)

A shared “YM2612 grammar” — with room for fingerprints.

Data Analysis: CON insights — wirings & musical reading

YM2612 algorithms as wiring diagrams

Alg 4 — two FM voices summed



Alg 2 — stacked / intertwined



Alg 5 — one mod → many carriers



Alg 0 — full serial stack

1 → 2 → 3 → 4



Gold $\approx$ carriers, cyan $\approx$ modulators.

- ▶ **CON 4** — two FM voices summed: bright + body, mix-friendly default
- ▶ **CON 5** — one mod → many carriers: bright, dense, multi-component
- ▶ **CON 2** — stacked / intertwined: richer, less stable spectra
- ▶ **CON 0** — full serial (1→2→3→4): metallic edge; powerful, harder to control

Data Analysis: Feedback as a binary aesthetic

YM2612 feedback (FL) on operator 1

FL = 0 — clean Op1



No self-loop: Op1 stays a
stable sine-like source

FL = 7 — Op1 self-feeds



Self-modulation: Op1 becomes a
harmonic / noise generator

Same idea as the CON wirings: Op 1 loop = FL.

Almost no middle ground: **FL = 0** or **FL = 7**.

- ▶ **FL = 7**: Op 1 becomes a harmonic / noise generator — grit, metallic cut for leads, basses, attacks
- ▶ **FL = 0**: stable sine-like op — clean carrier or precise modulator
- ▶ A **stylistic commitment** between two worlds, not a gentle trim



Data Analysis: Each CON is its own preset line — FL first

Changing CON rewires the instrument: each algorithm is a different preset family.
First lens: how feedback (FL) is used inside each CON.

Feedback (FL) by CON (role-aware)



- ▶ **CON 7** (four parallel carriers): **FL = 0** dominates; FL 1–7 stay low and flat — four clean sines; self-modulation rarely wanted
- ▶ **CON 0** (serial stack): strong **FL = 7** and a tall **FL = 0** — already heavy op→op modulation; Op 1 feedback is optional grit, not required
- ▶ **CON 6** (one stack + three carriers): same bipolar habit — with $FL \approx 0$ it can feel close to CON 7's additive world
- ▶ Most other CONs: classic corpus pattern — **FL = 7** as the loud default

Data Analysis: Feature engineering — Brightness & Complexity

Fifty-eight chip params are hard to narrate. We add two derived scores (0–1) to simplify comparisons across presets, games, and composers.

Brightness_Index

Combines (normalized):

- ▶ **TL** (40%) — louder ops $\Rightarrow$ brighter (TL inverted)
- ▶ **AR** (30%) — faster attacks
- ▶ **MUL** (30%) — higher harmonics

Averaged across the four operators.

Complexity_Score

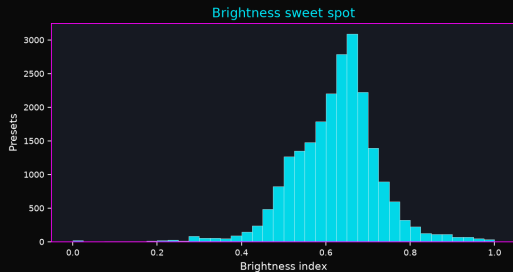
Combines (normalized):

- ▶ **CON** (40%) — algorithm index / routing
- ▶ **Active ops** (40%) — ops with TL < 127
- ▶ **FL** (20%) — feedback adds grit / motion

Proxy for internal FM motion, not audio “CPU cost.”

Used later for game / composer / nationality summaries and the embedding coloured maps.

Data Analysis: Brightness & Complexity — corpus insights



Brightness_Index clusters mid-high ($\approx 0.55\text{--}0.7$):

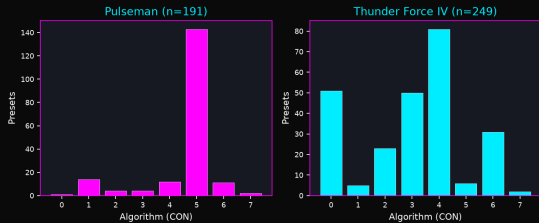
- ▶ Rarely dark by default — clarity for small TVs / dense mixes
- ▶ Not maxed at 1.0 — a **controlled sweet spot**, not pure harshness

Complexity_Score sits upper-mid ($\approx 0.6\text{--}0.85$):

- ▶ Multiple active ops + moderate routing — rich internal motion
- ▶ Very simple “almost sine” patches are rare
- ▶ Absolute max complexity avoided — depth with predictable playback

Data Analysis: Game fingerprints — commit vs variety

Shared YM2612 grammar — but each score builds a recognizable algorithm fingerprint.



- ▶ **Pulseman:** Alg. 5 dominates ($\approx 75\%$) — brand by *commitment* (techno bite)
- ▶ **Gunstar Heroes, Streets of Rage 3, Thunder Force IV:** spread across CONs — identity from *variety* / deliberate routing per patch
- ▶ Alg. 4 still rises often — the dependable default — but house style is how you leave (or stay in) it

Hear the fingerprint (same MIDI note — real game presets)

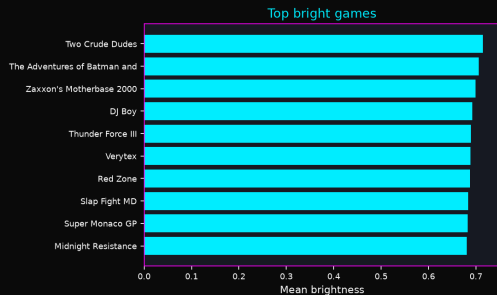
▶ **PM** Pulseman Alg.5

▶ **TF4** TF IV Alg.0

▶ **GH** Gunstar Alg.5

▶ **SoR3** SoR3 Alg.2

Data Analysis: Brightness by game — a shared sweet spot



Top titles cluster mid-high — enough bite for small TVs / dense mixes, not so bright it fatigues.

- ▶ **Bright end:** *The Adventures of Batman and Robin*, *Pulseman* — sharp, punchy energy
- ▶ **Rounder:** *Ristar*, *Sonic 3 & Knuckles* — still clear, more melodic body
- ▶ Nobody in this top set lives in the “dark” zone — a practical Genesis default

Bright vs rounder (same note)

▶ **BaR** Batman high B

▶ **PM** Pulseman high B

▶ **Rst** Ristar

▶ **S3K** Sonic 3&K

Data Analysis: Dark outliers — choosing warmth over bite



A small cluster deliberately sits at the **low** Brightness_Index — breaking the Genesis “cut through the TV” habit.

- ▶ Chart extremes: *Spider-Man & X-Men, Tinhead, The Tick, Thunder Pro Wrestling...*
- ▶ Warmth / weight / atmosphere over immediacy — often racing, sports, comedy pacing
- ▶ Likely lower TL, slower AR, fewer high MULs — room for groove, bass, PCM drums

High brightness was the default; rejecting it was a mood / genre choice.

Hear the dark side (same note — contrast with BaR/PM)

▶ **SMX** Spider-Man

▶ **Tick** The Tick

▶ **Tin** Tinhead

▶ **TG2** Top Gear 2

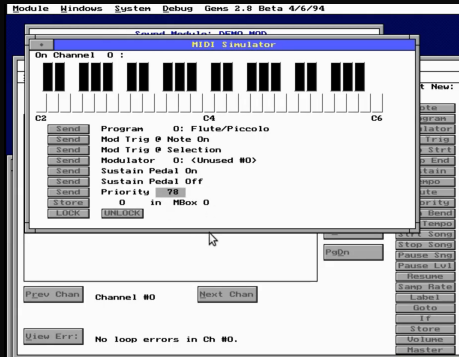
Data Analysis: GEMS — the American sound driver

GEMS (Genesis Editor for Music and Sound):
Sega of America's mid-1990s music tool.

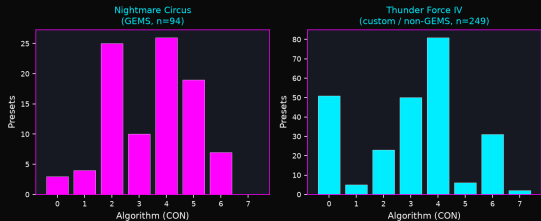
In our corpus it is a minority — but not a footnote:

- ▶ Only **7.3%** of presets (**43** / 547 games) — yet **1,600+** patches to study
- ▶ Big GEMS libraries: *Nightmare Circus*, *Marsupilami*, *Shadowrun*, *Chakan*
- ▶ Slightly *fewer* presets per game than non-GEMS — structured consistency over volume

Tool variable / industrial conditions — not a moral category.



Data Analysis: GEMS — a different FM grammar



Synthesis contrast (GEMS vs the wider Genesis corpus):

- ▶ Modal algorithm: **GEMS** → **Alg. 2**; ecosystem default **Alg. 4**
- ▶ Feedback: **GEMS** → **FL = 0**; non-GEMS almost always **FL = 7**
- ▶ Brightness / complexity only slightly lower — not “duller,” but **more controlled**

Stability and multi-track reliability over feedback-driven grit.

Hear the tool gap (same MIDI note)

▶ **NC** Nightmare Circus GEMS

▶ **TF4** TF IV non-GEMS

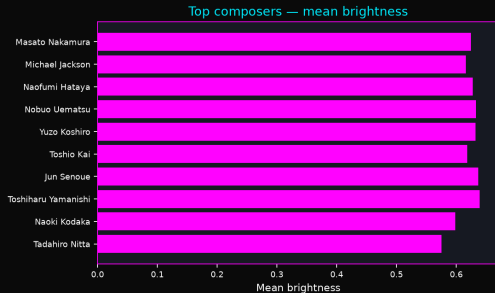
Data Analysis: Composer signatures — legends

Why composers matter: same chip, different sonic signatures — technical habits that make a voice recognisable.

- ▶ **Yuzo Koshiro** (*Streets of Rage*): house / techno into FM; programmed on hardware — high control, aggressive presence
- ▶ **Masato Nakamura** (*Sonic*): J-pop craft on chip limits — catchy beds that survive hundreds of loops
- ▶ **Hiroshi Kawaguchi** (*Golden Axe*, OutRun): arcade clarity — balance and envelopes that cut through noise
- ▶ **Norio Hanzawa** (*Gunstar Heroes*): Treasure experimentation — layered routing, unusual topologies

Proxy in the dump: credited games as composer neighbourhoods (metadata gaps remain).

Data Analysis: Composer signatures — shared FM grammar



Top composers converge on the same practical toolbox:

- ▶ Modal algorithm: almost everyone → **Alg. 4** (two-voice workhorse)
- ▶ Modal feedback: virtually all → **FL = 7** (default grit, not a special FX)
- ▶ Brightness means cluster $\approx 0.55\text{--}0.7$ — TV speakers + dense mixes
- ▶ Complexity means $\approx 0.65\text{--}0.8$ — controlled richness, not chaos

Style shows up in envelopes and game palettes — not in rejecting the chip defaults.

Data Analysis: Composer signatures — hear four voices

Same MIDI note — game-proxied presets from each neighbourhood.

Click to hear

▶ **YK** Koshiro SoR Alg.4 FL7

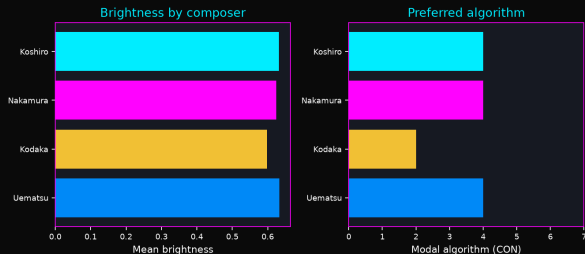
▶ **MN** Nakamura Sonic Alg.4 FL7

▶ **HK** Kawaguchi Golden Axe

▶ **NH** Hanzawa Gunstar

- ▶ **Koshiro**: brighter bite (Moon Beach) — club energy as a single voice
- ▶ **Nakamura**: Star Light bed — pop clarity inside the same Alg. 4 / FL = 7 grammar
- ▶ **Kawaguchi / Hanzawa**: Alg. 5 colour — arcade cut vs layered Treasure texture

Data Analysis: More signatures — Kodaka & Uematsu



- ▶ **Naoki Kodaka**: modal CON leans to **Alg. 2** — the clearest routing outlier among top names; darker / more experimental worlds
- ▶ **Nobuo Uematsu**: stays on the Alg. 4 workhorse, with slightly *lusher* complexity — controlled richness, not a different topology

Shared toolbox $\neq$ identical personality.

Regionally, Japan / USA / UK also share Alg. 4 + FL = 7; nationality barely separates embeddings — chip constraints dominate.

Hear the Kodaka lean (Alg. 2, FL = 7 — Super Fantasy Zone)

▶ **NK Kodaka Alg.2**

DAFMExplorer — three tools, one pipeline

We never ask the data for genre labels. We only ask: which presets look alike in chip-parameter space?

1. Compress (PCA)

~58 knobs per preset is a crowded room: many coordinates move together (e.g. operator levels).

Principal Component Analysis finds a smaller set of axes that keep most of the variance — roughly 58→50 dimensions — so later steps see signal, not redundant noise.

2. Group (K-Means)

In that compressed space, K-Means places K centres and assigns each preset to the nearest one.

We sweep K and pick the elbow ($K=7$): enough neighbourhoods to be useful, not so many that groups fragment into noise.

3. Lay out (UMAP)

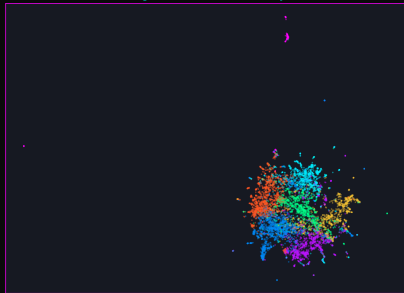
Humans cannot stare at 50D. UMAP folds the cloud onto a plane while trying to keep neighbours near neighbours.

That 2D layout is the clickable map — visualisation of structure already found, not a second clustering.

Unsupervised throughout: no composer / game / genre labels in the fit — only YM2612 parameters. Names come later, by listening.

DAFMExplorer — sonic realms & the live map

Timbral neighbourhoods (coloured by cluster)



Seven emergent groups → human “sonic realm” names (a reading aid, not a taxonomy we imposed first):

- ▶ Raw Signals
- ▶ Neon Action
- ▶ Polished Arcade
- ▶ High-Speed Chiptune
- ▶ Deep Space FM
- ▶ Fantasy Atmospheres
- ▶ Experimental Playgrounds

Same pipeline feeds nearest-neighbour “similar presets” in the app — still in parameter space.

Listen & explore

DAFMExplorer — map, real-time YM2612 playback, filters by realm / game / composer.

<https://dafm-explorer.vercel.app/>

Conclusions — what the study shows

- ▶ Shared YM2612 grammar: Alg. 4 and FL = 7 dominate as practical defaults across composers and regions
- ▶ Feature engineering (Brightness / Complexity) makes corpus-scale comparison discussable without drowning in 58 knobs
- ▶ Games leave fingerprints: some commit to one wiring; others keep a multi-algorithm palette
- ▶ Tools matter: GEMS leans Alg. 2 + FL = 0 — industrial conditions, not “worse FM”
- ▶ Composer signatures sit inside the shared toolbox; outliers (e.g. Kodaka Alg. 2) are readable in the data
- ▶ Unsupervised map organises ~93k presets into listen-able realms — the scaffold for DAFMExplorer

Presets are evidence of shared timbral practice on a platform — measurable, still audible, openly reusable.

Limits & future work — toward sound-based similarity

Limits (honest now):

- ▶ A preset $\neq$ a full score (melody, sequencing, PCM drums live elsewhere)
- ▶ Archive quirks; maps are exploratory; “national style” / “GEMS sound” stay hypotheses
- ▶ Today’s clustering / nearest neighbours use parameter space only — they do not hear the preset

Future work (in progress): render each patch $\rightarrow$ extract audio features $\rightarrow$ recommend by timbre, not only by knobs.

Related literature we aim to follow / adapt:

- ▶ Yee-King et al. (2018) — sound matching / auto-programming of FM (Dexed) from audio
- ▶ Cáceres (2007) — MFCC features + clustering for FM preset lookup
- ▶ Le Vaillant & Joder (2024) — timbre-regularised latent spaces over rendered presets
- ▶ Sound2Synth (IJCAI 2022) — estimating FM parameters from target sounds

Goal: a creation / imitation / exploration tool that advises presets similar in sound, not just in input parameters.

Thanks — questions welcome



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<https://www.kassersynths.com>

<https://www.tindie.com/stores/kassersynths/>

<https://dafm-explorer.vercel.app/>

<https://github.com/kassersynths/DAFMExplorer>

Thank you — Q&A